Q1a) Discuss about the mobility. What do you understand about the address migration?

Ans 1a) Mobility and Address Migration in Mobile Computing

Introduction

Mobile computing is a technology that allows users to communicate and access data while moving from one location to another using wireless networks and portable devices such as:

- Smartphones
- Laptops
- Tablets
- PDAs

One of the most important concepts in mobile computing is:

```
Mobility
```

Mobility enables continuous communication even when users change their physical location. To support mobility efficiently, techniques such as:

```
Address Migration
```

are used in mobile networks.

Mobility in Mobile Computing

Definition

Mobility refers to the ability of a user, device, or application to move freely across different geographical locations while maintaining uninterrupted communication and network connectivity.

In simple words:

- A mobile user can move from one network area to another without losing connection.

Characteristics of Mobility

1. User movement during communication.
2. Continuous connectivity.
3. Dynamic network changes.
4. Wireless communication support.
5. Location independence.

Types of Mobility

---

1. Terminal Mobility

Definition

Terminal mobility refers to the movement of a mobile device from one network location to another while maintaining communication.

Example

A smartphone moving from:

- Home Wi-Fi
  to
- Mobile network

during a video call.

---

2. Personal Mobility

Definition

Personal mobility allows users to access services using any device regardless of location.

Example

Accessing email from:

- Mobile phone
- Laptop
- Tablet

using same account.

---

3. Service Mobility

Definition

Service mobility allows users to maintain the same services while moving between networks.

Example

Continuing cloud storage access while traveling.

---

4. Session Mobility

Definition

Session mobility maintains an active communication session while the user changes devices or networks.

Example

Continuing a video meeting after switching from Wi-Fi to mobile data.

Importance of Mobility

Mobility is important because it:

1. Enables communication anywhere.
2. Supports real-time connectivity.
3. Improves flexibility.
4. Increases productivity.
5. Supports modern mobile applications.

Advantages of Mobility

1. Location independence.
2. Continuous communication.
3. Better accessibility.
4. Increased convenience.
5. Supports remote work and learning.

Challenges of Mobility

1. Network handoff issues.
2. Signal interruption.
3. Security concerns.
4. Limited battery power.
5. Address management complexity.

Address Migration in Mobile Computing

Introduction

When a mobile device moves from one network to another, its network address may change. Managing this address change without breaking communication is called:

```
Address Migration
```

Address migration is necessary to maintain continuous connectivity during mobility.

Definition of Address Migration

Address migration is the process of updating or transferring the network address of a mobile node when it moves from one network location to another.

It ensures:

- Ongoing communication continues smoothly.
- Packets reach the correct destination.

Why Address Migration is Needed

In traditional networks:

- IP addresses identify both:
    - Device identity
    - Device location

When a mobile device changes location:

- Its IP address may change.
- Existing communication sessions may break.

Address migration solves this problem.

Working of Address Migration

```
Mobile Device Moves
          ↓
New Network Detected
          ↓
New Address Assigned
          ↓
Address Updated
          ↓
Communication Continues
```

Components Involved in Address Migration

Component	Purpose
Mobile Node	Moving device
Home Network	Original network
Foreign Network	New network
Care-of Address	Temporary new address

Component	Purpose
Home Agent	Tracks mobile device

Address Migration in Mobile IP

Introduction

Mobile IP is a protocol designed to support mobility and address migration.

It allows mobile devices to:

- Move across networks
- Maintain permanent IP address

Important Terms in Mobile IP

---

1. Home Address

Definition

Permanent IP address assigned to the mobile device in its home network.

---

2. Care-of Address (CoA)

Definition

Temporary address assigned when the device enters a foreign network.

---

3. Home Agent

Definition

A router in the home network that tracks the mobile device’s current location.

---

4. Foreign Agent

Definition

A router in the foreign network that provides the care-of address.

Address Migration Process in Mobile IP

Step 1: Mobile Node Moves

The device moves from:

- Home network
  to
- Foreign network

Step 2: New Address Allocation

Foreign network assigns:

```
Care-of Address
```

Step 3: Registration

Mobile node registers new address with:

```
Home Agent
```

Step 4: Packet Forwarding

Home Agent forwards packets to:

- Care-of Address

Diagram of Address Migration

```
Home Network
     ↓
Mobile Node Moves
     ↓
Foreign Network
     ↓
Care-of Address Assigned
     ↓
Home Agent Updated
     ↓
Communication Continues
```

Advantages of Address Migration

1. Maintains ongoing communication.
2. Supports seamless mobility.
3. Reduces connection interruption.
4. Enables roaming across networks.
5. Improves user experience.

Problems in Address Migration

1. Increased signaling overhead.
2. Security vulnerabilities.
3. Packet delay during migration.
4. Handoff latency.
5. Complex routing management.

Applications of Address Migration

Application	Usage
Mobile Internet	Continuous browsing
Video Calling	Seamless communication
Online Gaming	Session continuity

Application	Usage
Mobile Banking	Secure connectivity
IoT Devices	Dynamic network movement

Difference Between Mobility and Address Migration

Aspect	Mobility	Address Migration
Definition	Movement of user/device	Change/update of network address
Purpose	Maintain connectivity	Maintain routing and communication
Focus	User movement	Address handling
Example	Moving during call	Assigning new IP address

Real-Life Example

Suppose a user is attending an online meeting while traveling:

- The phone moves from one cellular tower to another.
- Network changes occur.
- Address migration updates the network address.
- The meeting continues without interruption.

Advantages of Mobility in Mobile Computing

1. Communication anywhere and anytime.
2. Better flexibility.
3. Increased productivity.
4. Supports modern mobile applications.
5. Enables real-time access to services.

Conclusion

Mobility is a fundamental concept in mobile computing that allows users and devices to move freely while maintaining network connectivity. To support this movement efficiently, address migration techniques are used to update network addresses dynamically without interrupting communication. Technologies such as Mobile IP help achieve seamless mobility by using concepts like home address, care-of address, and home agents. Mobility and address migration together enable modern wireless communication systems to provide uninterrupted and efficient mobile services.

Q1b) What are the parameters considered for defining QoS in GPRS technology?

Ans 1b) Quality of Service (QoS) Parameters in GPRS Technology

Introduction

GPRS (General Packet Radio Service) is a packet-switched mobile data technology used in GSM networks for providing internet and multimedia services. It enables services such as:

- Web browsing
- Email
- Multimedia messaging
- Mobile applications

Different applications require different levels of network performance. For example:

- Video streaming requires high speed and low delay.
- Email requires less bandwidth.

To ensure proper network performance for different services, GPRS uses:

```
Quality of Service (QoS)
```

parameters.

Definition of QoS in GPRS

Quality of Service (QoS) in GPRS refers to the set of performance characteristics that define how efficiently and reliably data communication services are provided to users.

QoS ensures that network resources are allocated according to application requirements.

Importance of QoS in GPRS

QoS is important because it:

1. Improves user experience.
2. Ensures reliable communication.
3. Supports multimedia applications.
4. Manages network resources efficiently.
5. Prioritizes important services.

QoS Architecture in GPRS

```
User Application
        ↓
QoS Parameters
        ↓
GPRS Network
        ↓
Data Transmission
```

Parameters Considered for Defining QoS in GPRS

The main QoS parameters in GPRS are:

1. Precedence Class
2. Reliability Class
3. Delay Class
4. Throughput Class
5. Mean Throughput
6. Traffic Handling Priority

---

1. Precedence Class

Definition

Precedence Class defines the priority level assigned to a data packet or service in the GPRS network.

It determines which user or application receives higher priority during network congestion.

Types of Precedence Classes

Class	Priority Level
High Priority	Highest importance
Normal Priority	Standard services
Low Priority	Least important traffic

Example

- Emergency communication → High priority
- Email service → Normal priority
- Background downloads → Low priority

Importance

1. Helps manage network congestion.
2. Ensures critical services receive preference.
3. Improves network efficiency.

---

2. Reliability Class

Definition

Reliability Class defines the accuracy and reliability of data transmission in GPRS.

It specifies:

- Error probability

- Packet loss
- Data duplication

Reliability Factors

Factor	Description
Packet Loss	Lost data packets
Data Corruption	Incorrect data received
Packet Duplication	Duplicate packets

Classes of Reliability

Different classes provide different levels of:

- Error protection
- Data integrity

Example

Banking applications require:

```
High Reliability
```

to avoid data corruption.

Importance

1. Ensures accurate communication.
2. Reduces transmission errors.
3. Improves application reliability.

---

3. Delay Class

Definition

Delay Class defines the maximum acceptable delay for transmitting data packets between sender and receiver.

It measures:

- Transmission delay
- Network latency

Delay Sensitivity

Some applications require:

- Low delay
  while others can tolerate higher delay.

Example

Application	Delay Requirement
Video Call	Very Low Delay
Email	Higher Delay Acceptable

Delay Classes

Class	Delay Level
Class 1	Very Low Delay
Class 2	Moderate Delay
Class 3	Higher Delay

Importance

1. Improves real-time communication.
2. Essential for voice/video applications.
3. Enhances user experience.

---

4. Throughput Class

Definition

Throughput Class specifies the maximum data transfer rate supported by the network.

It defines:

- Upload speed
- Download speed

Measurement

Throughput is measured in:

```
bits per second (bps)
```

Example

Video streaming requires:

```
High Throughput
```

for smooth playback.

Importance

1. Determines network speed.
2. Supports multimedia applications.
3. Improves data transfer efficiency.

---

5. Mean Throughput

Definition

Mean Throughput represents the average data transfer rate achieved over a period of time.

Unlike throughput class:

- It measures average performance rather than maximum speed.

Example

A user may receive:

- Maximum speed: 100 kbps
- Average speed: 40 kbps

The average speed is the mean throughput.

Importance

1. Reflects actual network performance.
2. Helps evaluate service quality.
3. Important for network optimization.

---

6. Traffic Handling Priority

Definition

Traffic Handling Priority determines how packets are processed during heavy network traffic conditions.

It is mainly used for:

- Interactive applications
- Real-time communication

Priority Levels

Level	Description

Level	Description
High	Immediate processing
Medium	Standard processing
Low	Delayed processing

Example

Voice calls receive:

```
Higher Priority
```

than file downloads.

Importance

1. Ensures smooth multimedia services.
2. Improves real-time communication.
3. Reduces service interruption.

Additional QoS Considerations in GPRS

---

1. Packet Loss Rate

Measures percentage of lost packets during transmission.

---

2. Jitter

Represents variation in packet arrival time.

Important for:

- Audio streaming
- Video conferencing

---

3. Availability

Measures network uptime and service accessibility.

QoS Negotiation in GPRS

Introduction

When a mobile device connects to the GPRS network:

- QoS parameters are negotiated between:
    - Mobile station
    - Network

Process

```
User Requests Service
          ↓
QoS Parameters Sent
          ↓
Network Evaluates Resources
          ↓
QoS Profile Assigned
```

QoS Profiles in GPRS

Different applications receive different QoS profiles.

Application	QoS Requirement
Voice Call	Low delay
Email	High reliability
Video Streaming	High throughput
Banking	High reliability and security

Advantages of QoS in GPRS

1. Better user experience.
2. Efficient resource allocation.
3. Supports multimedia applications.
4. Improves network reliability.
5. Enables service differentiation.

Challenges in QoS Management

1. Limited bandwidth.
2. Network congestion.
3. Mobility management.
4. Wireless signal interference.
5. Dynamic traffic conditions.

Real-Life Examples

Application	Important QoS Parameter
YouTube Streaming	Throughput
WhatsApp Calling	Delay
Online Banking	Reliability
File Downloading	Mean Throughput

Comparison of QoS Parameters

Parameter	Purpose
Precedence Class	Defines priority
Reliability Class	Ensures data accuracy
Delay Class	Controls latency
Throughput Class	Defines maximum speed
Mean Throughput	Measures average speed
Traffic Handling Priority	Manages packet processing

Conclusion

Quality of Service (QoS) in GPRS is essential for providing efficient and reliable mobile data communication. QoS parameters such as precedence class, reliability class, delay class, throughput class, mean throughput, and traffic handling priority help manage network resources according to application requirements. These parameters ensure that different services such as video streaming, voice communication, and web browsing receive appropriate network performance. Proper QoS management improves user experience, enhances network efficiency, and supports modern multimedia mobile applications.

Q1c) Write point wise disadvantages of WLAN. Define cell splitting in mobile computing.

Ans 1c) Disadvantages of WLAN (Wireless Local Area Network)

Introduction

A WLAN (Wireless Local Area Network) is a wireless communication network that allows devices to connect and communicate using radio waves instead of physical cables.

Although WLAN provides mobility and flexibility, it also has several disadvantages.

Point-wise Disadvantages of WLAN

---

1. Security Issues

Wireless signals can be intercepted easily by unauthorized users.

This may lead to:

- Hacking
- Data theft
- Unauthorized access

---

2. Limited Range

WLAN has limited coverage area compared to wired networks.

Signal strength decreases with:

- Distance
- Walls
- Obstacles

---

3. Lower Speed Compared to Wired Networks

Wireless networks generally provide lower speed and stability than wired LAN connections.

Performance decreases when many users connect simultaneously.

---

4. Signal Interference

WLAN signals may suffer interference from:

- Microwave ovens
- Bluetooth devices
- Other wireless networks

This reduces network quality.

---

5. Reliability Problems

Wireless connections may disconnect due to:

- Weak signals
- Environmental conditions
- Device movement

---

6. High Power Consumption

Wireless communication consumes more battery power in mobile devices.

Continuous Wi-Fi usage drains battery faster.

---

7. Bandwidth Sharing

All users share the same wireless bandwidth.

As the number of connected devices increases:

- Network speed decreases.

---

8. Installation Cost

Initial setup cost can be high due to:

- Wireless routers
- Access points
- Security configurations

---

9. Health Concerns

Some people believe long-term exposure to radio frequency signals may affect health.

---

10. Difficulty in Troubleshooting

Wireless network issues are harder to identify compared to wired networks because of:

- Signal interference
- Dynamic connectivity

Definition of Cell Splitting in Mobile Computing

Introduction

Cell splitting is a technique used in cellular mobile communication systems to increase network capacity and improve service quality.

Definition

Cell Splitting is the process of dividing a large cell into smaller cells, each with its own base station and lower transmission power, to support more users in densely populated areas.

Purpose of Cell Splitting

1. Increase network capacity.
2. Reduce traffic congestion.
3. Improve signal quality.
4. Support more mobile users.

Working of Cell Splitting

Initially:

- A large cell covers a wide area.

When traffic increases:

- The large cell is divided into smaller cells.
- New base stations are installed.

Diagram of Cell Splitting

```
Before Splitting:
       Large Cell

          ○


After Splitting:
     ○    ○
       ○
     ○    ○
```

Advantages of Cell Splitting

1. Increased network capacity.
2. Better frequency reuse.
3. Improved coverage quality.
4. Reduced call blocking.

Disadvantages of Cell Splitting

1. Increased infrastructure cost.
2. More complex network management.
3. Increased handoff frequency.

Example

In crowded urban cities:

- Mobile traffic becomes very high.
- Telecom operators divide large cells into smaller cells to manage more users efficiently.

Conclusion

WLAN provides wireless connectivity and mobility but suffers from disadvantages such as security issues, limited range, interference, and reduced speed. Cell splitting is an important technique in mobile computing used to improve cellular network capacity by dividing large cells into smaller ones, thereby supporting more users and enhancing communication quality.

Q2a) Explain HLR and VLR concept for location management in mobile communication.

Ans 2a) HLR and VLR in Mobile Communication

Introduction

In mobile communication systems such as GSM, users continuously move from one location to another while remaining connected to the network. To provide uninterrupted communication, the network must always know the current location of the mobile user.

This process is called:

```
Location Management
```

Two important databases used for location management are:

1. HLR (Home Location Register)
2. VLR (Visitor Location Register)

These databases help track mobile subscribers and provide seamless communication services.

Location Management in Mobile Communication

Definition

Location management is the process of tracking and updating the current location of a mobile subscriber in a cellular network.

It helps the network:

- Deliver calls
- Send messages
- Provide roaming services

efficiently.

Need for Location Management

Location management is necessary because:

1. Mobile users move frequently.
2. Calls and messages must reach the correct location.

3. Roaming users need network access.
4. Efficient resource management is required.

HLR (Home Location Register)

Definition

HLR (Home Location Register) is a central database in a mobile network that permanently stores information about all subscribers belonging to a specific network operator.

It contains:

- Subscriber details
- Service information
- Current location information

Characteristics of HLR

1. Permanent database.
2. Stores subscriber profile.
3. Maintains current VLR location.
4. One HLR serves many subscribers.

Information Stored in HLR

Information	Description
Subscriber Identity	IMSI number
Mobile Number	MSISDN
Service Subscription	Enabled services
Current VLR Address	Current location information
Authentication Data	Security information

Functions of HLR

1. Stores permanent subscriber data.

2. Maintains user service details.
3. Tracks subscriber location.
4. Supports call routing.
5. Provides roaming support.

Example of HLR Working

Suppose a user belongs to:

```
Airtel Delhi Network
```

The user's permanent information is stored in the HLR of Airtel Delhi.

Even if the user travels to another city:

- HLR still stores subscriber details permanently.

Advantages of HLR

1. Centralized subscriber management.
2. Supports roaming.
3. Improves network efficiency.
4. Maintains permanent subscriber records.

VLR (Visitor Location Register)

Definition

VLR (Visitor Location Register) is a temporary database that stores information about subscribers currently present in a particular area served by a Mobile Switching Center (MSC).

It stores temporary information of visiting users.

Characteristics of VLR

1. Temporary database.

2. Stores roaming subscriber information.
3. Connected to MSC.
4. Data removed when user leaves area.

Information Stored in VLR

Information	Description
Temporary Subscriber Data	Current user details
Location Area Information	User's current area
Authentication Information	Security data
Service Permissions	Allowed services

Functions of VLR

1. Tracks roaming subscribers.
2. Reduces HLR query traffic.
3. Supports call delivery.
4. Stores temporary user information.
5. Speeds up location management.

Example of VLR Working

Suppose a user from Delhi travels to Mumbai.

Then:

- Mumbai network creates a temporary VLR entry.
- HLR is updated with the current VLR address.

Now calls are routed through the Mumbai VLR.

Working of HLR and VLR Together

Step 1: Subscriber Moves to New Area

The mobile user enters a new location area.

Step 2: VLR Registration

The local VLR stores temporary subscriber details.

Step 3: HLR Update

VLR informs the HLR about the subscriber's new location.

Step 4: Call Routing

When someone calls the subscriber:

- HLR checks current VLR location.
- Call is forwarded through the correct VLR.

Diagram of HLR and VLR Operation

```
Mobile User
       ↓
New Location Area
       ↓
VLR Stores Temporary Data
       ↓
HLR Updated
       ↓
Call Routed Correctly
```

Comparison Between HLR and VLR

Feature	HLR	VLR
Full Form	Home Location Register	Visitor Location Register
Type	Permanent Database	Temporary Database
Data Stored	Permanent subscriber data	Temporary roaming data

Feature	HLR	VLR
Location	Centralized	Associated with MSC
Purpose	Subscriber management	Visitor tracking
Data Duration	Permanent	Temporary

Role of HLR and VLR in Roaming

When users travel outside their home network:

- HLR maintains permanent information.
- VLR temporarily handles the subscriber locally.

This allows:

- Seamless roaming
- Continuous communication

Advantages of Using HLR and VLR

1. Efficient location tracking.
2. Faster call setup.
3. Reduced network congestion.
4. Supports roaming services.
5. Improves mobile communication reliability.

Challenges in HLR and VLR Management

1. Database synchronization complexity.
2. Security threats.
3. Signaling overhead.
4. Handling large subscriber volumes.

Real-Life Example

Suppose a user from Delhi travels to Bangalore.

Process:

1. Bangalore VLR registers the user.
2. Delhi HLR updates user location.
3. Incoming calls are forwarded to Bangalore network.
4. User continues communication seamlessly.

Importance in Mobile Communication

HLR and VLR are essential because they:

- Maintain mobility support.
- Enable location tracking.
- Improve communication efficiency.
- Support nationwide roaming.

Conclusion

HLR (Home Location Register) and VLR (Visitor Location Register) are important databases used in mobile communication systems for location management. HLR stores permanent subscriber information, while VLR temporarily stores information about roaming users in a particular area. Together, they enable seamless mobility, efficient call routing, and uninterrupted communication in cellular networks. These concepts are fundamental to the successful operation of GSM and other mobile communication systems.

Q2b) Discuss about TCP/IP protocol suite with suitable example of protocol defined at each and
every layer.

Ans 2b) TCP/IP Protocol Suite

Introduction

The TCP/IP Protocol Suite is the standard communication model used for data transmission over computer networks and the Internet. It is a collection of communication protocols that enables devices to communicate with each other reliably.

TCP/IP stands for:

`Transmission Control Protocol / Internet Protocol`

It was developed by the:

- U.S. Department of Defense (DoD)

and is now the foundation of:

- Internet communication
- Mobile communication
- Wireless networking

Definition of TCP/IP Protocol Suite

The TCP/IP protocol suite is a set of layered communication protocols used for transmitting data between interconnected devices over networks.

It defines:

- How data is transmitted
- How devices are addressed
- How routing occurs
- How communication is managed

Features of TCP/IP Protocol Suite

1. Open standard protocol.
2. Supports heterogeneous networks.
3. Reliable communication.
4. Scalable architecture.
5. Supports routing and addressing.

Layers of TCP/IP Protocol Suite

The TCP/IP model consists of four layers:

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Network Access Layer

TCP/IP Layered Architecture

```
Application Layer
        ↓
Transport Layer
        ↓
Internet Layer
        ↓
Network Access Layer
```

---

1. Application Layer

Introduction

The Application Layer provides services directly to user applications and enables communication between software applications.

It combines the functions of:

- Application layer
- Presentation layer
- Session layer

of the OSI model.

Functions of Application Layer

1. File transfer
2. Email communication
3. Web browsing
4. Remote login
5. Name resolution

Protocols at Application Layer

Protocol	Purpose
HTTP	Web browsing
HTTPS	Secure web communication
FTP	File transfer

Protocol	Purpose
SMTP	Sending emails
POP3	Receiving emails
DNS	Domain name resolution
Telnet	Remote login

Example of Application Layer Protocol

HTTP (HyperText Transfer Protocol)

Used for:

- Accessing websites

Example:

```
https://www.google.com
```

The browser uses HTTP/HTTPS to communicate with web servers.

---

2. Transport Layer

Introduction

The Transport Layer provides:

- End-to-end communication
- Data reliability
- Flow control
- Error handling

It ensures data reaches the correct application process.

Functions of Transport Layer

1. Data segmentation
2. Error detection
3. Flow control
4. Reliable transmission

5. Multiplexing

Protocols at Transport Layer

Protocol	Purpose
TCP	Reliable communication
UDP	Fast communication

---

a) TCP (Transmission Control Protocol)

Features

1. Connection-oriented.
2. Reliable transmission.
3. Error checking.
4. Ordered delivery.

Example of TCP Usage

Used in:

- Web browsing
- Email
- Online banking

where reliable communication is necessary.

---

b) UDP (User Datagram Protocol)

Features

1. Connectionless.
2. Faster communication.
3. No guarantee of delivery.

Example of UDP Usage

Used in:

- Video streaming
- Online gaming
- Voice calls

where speed is more important than reliability.

Example of Transport Layer

Suppose a user downloads a file:

- TCP ensures all packets arrive correctly and in order.

---

3. Internet Layer

Introduction

The Internet Layer is responsible for:

- Logical addressing
- Routing
- Packet forwarding

It determines the path data takes across networks.

Functions of Internet Layer

1. IP addressing
2. Routing packets
3. Fragmentation
4. Packet delivery

Protocols at Internet Layer

Protocol	Purpose
IP	Addressing and routing
ICMP	Error reporting
ARP	Address resolution
RARP	Reverse address mapping

---

a) IP (Internet Protocol)

Purpose

Provides logical addressing using:

```
IP Addresses
```

Example

```
192.168.1.1
```

---

b) ICMP (Internet Control Message Protocol)

Purpose

Used for:

- Error reporting
- Network diagnostics

Example

```
Ping Command
```

uses ICMP packets.

---

c) ARP (Address Resolution Protocol)

Purpose

Converts:

- IP address
  to
- MAC address

Example

Finding hardware address of a destination device.

Example of Internet Layer

When a user sends data to another network:

- IP protocol determines the route for packet delivery.

---

4. Network Access Layer

Introduction

The Network Access Layer handles:

- Physical transmission of data
- Hardware addressing
- Communication with network devices

It combines:

- Physical layer
- Data Link layer

of the OSI model.

Functions of Network Access Layer

1. Physical data transmission.
2. Frame creation.
3. MAC addressing.
4. Error detection.

Protocols at Network Access Layer

Protocol	Purpose
Ethernet	Wired communication
Wi-Fi	Wireless communication
PPP	Point-to-point communication
Frame Relay	WAN communication

Example of Network Access Layer

Ethernet

Used in:

- LAN networks
- Office networks

for wired communication.

Data Encapsulation in TCP/IP

As data moves through layers:

- Each layer adds its own header.

Encapsulation Process

```
Application Data
        ↓
TCP Segment
        ↓
IP Packet
        ↓
Frame
        ↓
Transmission Medium
```

Working Example of TCP/IP Protocol Suite

Example: Opening a Website

Suppose a user opens:

```
www.google.com
```

Step 1: Application Layer

- HTTP request generated.
- DNS resolves website address.

Step 2: Transport Layer

- TCP establishes reliable connection.

Step 3: Internet Layer

- IP routes packets to destination server.

Step 4: Network Access Layer

- Ethernet/Wi-Fi transmits data physically.

Advantages of TCP/IP Protocol Suite

1. Reliable communication.
2. Scalable architecture.
3. Supports heterogeneous networks.
4. Widely accepted standard.
5. Supports routing across Internet.

Disadvantages of TCP/IP Protocol Suite

1. Complex configuration.
2. Security issues.
3. No guaranteed QoS.
4. Large header overhead.

Comparison Between TCP/IP and OSI Model

Feature	TCP/IP Model	OSI Model
Number of Layers	4	7
Practical Usage	Widely used	Mainly reference model
Development	DoD	ISO
Protocol Dependency	Protocol-based	Generic model

Real-Life Applications of TCP/IP

Application	Protocol Used
Web Browsing	HTTP + TCP + IP
Email	SMTP + TCP + IP
Video Streaming	UDP + IP
File Transfer	FTP + TCP + IP

Importance of TCP/IP in Mobile Computing

TCP/IP enables:

- Mobile internet
- Wireless communication
- Cloud services
- Mobile application networking

It is the backbone of:

- 4G/5G communication
- Wi-Fi networks
- Internet services

Conclusion

The TCP/IP protocol suite is the foundation of modern networking and Internet communication. It consists of four layers: Application Layer, Transport Layer, Internet Layer, and Network Access Layer. Each layer performs specific functions and uses different protocols such as HTTP, TCP, IP, and Ethernet. The layered architecture ensures reliable, scalable, and efficient communication between devices across networks. TCP/IP plays a crucial role in mobile computing, wireless communication, and global Internet connectivity.

Q2c) Define transmission impediments and its types in mobile computing? And also explain Doppler
Effect in details.

Ans 2c) Transmission Impediments and Doppler Effect in Mobile Computing

Introduction

In mobile computing and wireless communication, data is transmitted through wireless channels using electromagnetic waves. During transmission, signals may get weakened, distorted, or interfered with due to various environmental and physical factors.

These problems are known as:

```
Transmission Impediments
```

Transmission impediments reduce communication quality and may lead to:

- Data loss
- Signal distortion
- Reduced speed
- Call drops

One important phenomenon related to wireless communication is:

```
Doppler Effect
```

which occurs because of the relative movement between transmitter and receiver.

Transmission Impediments

Definition

Transmission impediments are factors that degrade or affect the quality of signal transmission in a communication system.

These impediments reduce the effectiveness of wireless communication and create transmission errors.

Causes of Transmission Impediments

1. Distance between sender and receiver.
2. Environmental obstacles.
3. Electromagnetic interference.
4. Mobility of users.
5. Atmospheric conditions.

Types of Transmission Impediments

The major transmission impediments in mobile computing are:

1. Attenuation
2. Distortion
3. Noise
4. Fading
5. Interference

---

1. Attenuation

Definition

Attenuation is the gradual loss of signal strength as the signal travels through a transmission medium.

As distance increases:

- Signal power decreases.

Causes of Attenuation

1. Long transmission distance.
2. Obstacles such as buildings.
3. Absorption by atmosphere.

Effects of Attenuation

- Weak signals
- Reduced communication quality
- Call drops

Example

Mobile signal becomes weaker when moving far from a cellular tower.

Diagram of Attenuation

```
Strong Signal ──────────→ Weak Signal
```

Solutions

1. Use repeaters/amplifiers.
2. Increase transmission power.
3. Use efficient antennas.

---

2. Distortion

Definition

Distortion occurs when different frequency components of a signal travel at different speeds, changing the original shape of the signal.

Causes of Distortion

1. Different propagation delays.
2. Multipath transmission.
3. Frequency variations.

Effects of Distortion

- Incorrect data reception
- Reduced signal quality

Example

Audio signal becoming unclear during a phone call.

Solutions

1. Equalization techniques.
2. Signal processing methods.

---

3. Noise

Definition

Noise is any unwanted electrical or electromagnetic signal that interferes with the original transmitted signal.

Types of Noise

Type	Description
Thermal Noise	Due to heat
Intermodulation Noise	Mixing of signals
Crosstalk	Interference from nearby channels
Impulse Noise	Sudden disturbances

Effects of Noise

1. Data corruption.
2. Communication errors.
3. Reduced signal clarity.

Example

Static sound during radio communication.

Solutions

1. Error detection techniques.
2. Shielded cables.
3. Noise filtering.

---

4. Fading

Definition

Fading is the variation in signal strength caused by changes in the transmission environment.

Causes of Fading

1. User mobility.
2. Multipath propagation.
3. Weather conditions.

Types of Fading

Type	Description
Slow Fading	Gradual signal variation
Fast Fading	Rapid signal fluctuation

Effects of Fading

- Signal fluctuation
- Temporary communication loss

Example

Mobile signal fluctuating while traveling in a vehicle.

Solutions

1. Diversity techniques.
2. Adaptive modulation.

---

5. Interference

Definition

Interference occurs when unwanted external signals affect the original communication signal.

Types of Interference

Type	Description
Co-channel Interference	Same frequency interference
Adjacent Channel Interference	Nearby frequency overlap

Effects

1. Poor call quality.
2. Reduced network performance.

Example

Two nearby wireless routers causing Wi-Fi interference.

Solutions

1. Proper frequency allocation.
2. Channel separation techniques.

Summary of Transmission Impediments

Impediment	Main Effect
Attenuation	Signal weakening
Distortion	Signal shape change
Noise	Unwanted signals
Fading	Signal fluctuation
Interference	External signal disturbance

Doppler Effect in Mobile Computing

Introduction

The Doppler Effect is an important phenomenon in wireless and mobile communication systems.

It occurs because:

- Mobile users move continuously.
- Relative motion exists between transmitter and receiver.

This movement changes the observed frequency of the received signal.

Definition of Doppler Effect

The Doppler Effect is the change in frequency of a wave observed due to the relative motion between the transmitter and the receiver.

Basic Concept

- If transmitter and receiver move closer:
    - Frequency increases.
- If transmitter and receiver move apart:

    - Frequency decreases.

Real-Life Example

The sound of an ambulance siren:

- Appears higher when approaching.
- Appears lower when moving away.

Similar behavior occurs in wireless signals.

Doppler Effect in Wireless Communication

In mobile communication:

- Mobile users move at different speeds.
- Signal frequency shifts because of movement.

This frequency shift is called:

```
Doppler Shift
```

Doppler Shift Formula

```
fd = (v / λ) cos θ
```

Where:

Symbol	Meaning
fd	Doppler shift frequency
v	Velocity of mobile user
λ	Wavelength
θ	Angle of motion

Working of Doppler Effect

```
Moving Receiver
        ↓
Frequency Changes
        ↓
Signal Variation Occurs
```

Types of Doppler Effect

---

1. Positive Doppler Shift

Definition

Occurs when transmitter and receiver move toward each other.

Result

Observed frequency:

```
Increases
```

---

2. Negative Doppler Shift

Definition

Occurs when transmitter and receiver move away from each other.

Result

Observed frequency:

```
Decreases
```

Causes of Doppler Effect in Mobile Computing

1. Vehicle movement.
2. User mobility.
3. Satellite communication.
4. Fast-moving wireless devices.

Effects of Doppler Effect

1. Frequency shift.
2. Signal fading.
3. Communication errors.
4. Reduced signal quality.

Doppler Effect and Fading

Doppler Effect contributes to:

```
Fast Fading
```

because rapid movement causes continuous frequency changes.

Applications of Doppler Effect

Application	Usage
Radar Systems	Speed detection
Satellite Communication	Frequency adjustment
Mobile Networks	Mobility analysis
GPS Systems	Position tracking

Advantages of Doppler Effect

1. Helps measure speed.

2. Useful in radar technology.
3. Supports mobility analysis.

Disadvantages of Doppler Effect

1. Causes frequency distortion.
2. Reduces communication quality.
3. Creates signal fading.

Methods to Reduce Doppler Effect

1. Adaptive equalization.
2. Frequency correction algorithms.
3. Diversity techniques.
4. Error correction methods.

Difference Between Transmission Impediments and Doppler Effect

Aspect	Transmission Impediments	Doppler Effect
Definition	Signal degradation factors	Frequency change due to motion
Cause	Noise, fading, attenuation	Relative movement
Effect	Communication quality reduction	Frequency shift

Real-Life Example

Suppose a user is traveling in a fast-moving train while talking on a mobile phone:

- Signal strength changes continuously.
- Frequency shifts occur because of movement.
- Doppler Effect causes fading and communication disturbances.

Conclusion

Transmission impediments are factors that negatively affect signal transmission in mobile computing systems. Major impediments include attenuation, distortion, noise, fading, and interference. These issues reduce communication quality and network performance. The Doppler Effect is a special phenomenon caused by relative motion between transmitter and receiver, resulting in frequency shifts in wireless signals. Understanding transmission impediments and Doppler Effect is essential for designing efficient and reliable mobile communication systems.

Q3a) a. Why does power control become one of the main issues for the efficient operation of CDMA?

Ans3a) Power Control in CDMA and Its Importance

Introduction

CDMA (Code Division Multiple Access) is a multiple access technique used in mobile communication systems where multiple users share the same frequency band simultaneously using unique spreading codes.

Unlike FDMA and TDMA:

- All users in CDMA transmit over the same frequency at the same time.

Because of this shared communication medium, efficient:

```
Power Control
```

becomes extremely important for proper system operation.

Without proper power control:

- Strong signals may dominate weak signals.
- Communication quality may degrade.
- Network capacity may reduce significantly.

Therefore, power control is considered one of the most critical issues in CDMA systems.

What is Power Control in CDMA?

Definition

Power control is the process of adjusting the transmission power of mobile users so that signals arrive at the base station with nearly equal power levels.

Its main objective is:

- To minimize interference.
- To maintain communication quality.

Need for Power Control in CDMA

In CDMA:

- All users use the same frequency channel simultaneously.
- Signals are separated using unique codes.

If one user transmits with very high power:

- It creates strong interference for other users.

This problem is known as:

```
Near-Far Problem
```

Hence, power control is essential.

Near-Far Problem in CDMA

Definition

The Near-Far Problem occurs when a mobile user close to the base station transmits a much stronger signal than a user located far away.

As a result:

- Strong signals overpower weak signals.
- Weak user communication may fail.

Example of Near-Far Problem

Suppose:

- User A is near the base station.
- User B is far from the base station.

If both transmit with same power:

- Signal from User A reaches the base station much stronger.
- User B's signal gets buried under interference.

Diagram of Near-Far Problem

```
User A (Near BTS) ─── Strong Signal
User B (Far BTS) ─── Weak Signal

Strong signal interferes with weak signal.
```

Why Power Control is a Main Issue in CDMA

Power control becomes a major issue because of the following reasons:

---

1. All Users Share Same Frequency Band

In CDMA:

- Every user transmits simultaneously over the same channel.

Therefore:

- Each user's signal acts as interference for others.

Improper power levels increase:

- Multiple Access Interference (MAI)

Result

Network performance decreases significantly.

---

2. Reduces Near-Far Effect

Power control ensures:

- Signals from all users reach the base station with nearly equal strength.

This prevents:

- Strong users from dominating weaker users.

Importance

Without power control:

- Weak users may lose connectivity.

---

3. Improves Signal Quality

Proper power adjustment improves:

- Signal-to-Interference Ratio (SIR)
- Signal-to-Noise Ratio (SNR)

Result

1. Better voice quality.
2. Fewer call drops.
3. Reliable communication.

---

4. Increases System Capacity

In CDMA:

- Capacity depends heavily on interference levels.

Higher interference means:

- Fewer users can communicate simultaneously.

Power control reduces interference and allows:

- More users per cell.

Importance

Efficient power control directly increases:

```
Network Capacity
```

---

5. Conserves Battery Power

Mobile devices operate on limited battery power.

Power control reduces unnecessary transmission power.

Benefits

1. Longer battery life.
2. Reduced heat generation.
3. Energy-efficient communication.

---

6. Minimizes Interference

Excess transmission power creates:

- Inter-cell interference
- Co-channel interference

Power control helps maintain optimal signal strength.

Result

Better overall network performance.

---

7. Supports Mobility

Mobile users continuously move:

- Toward or away from base stations.

Signal strength changes dynamically.

Power control continuously adjusts transmission power according to:

- User distance
- Signal conditions

Types of Power Control in CDMA

There are mainly two types of power control:

1. Open Loop Power Control
2. Closed Loop Power Control

---

1. Open Loop Power Control

Definition

In open loop power control:

- Mobile station estimates signal strength itself.
- Adjusts transmission power accordingly.

Working

```
Measure Received Signal
          ↓
Estimate Path Loss
          ↓
Adjust Transmission Power
```

Advantages

1. Simple implementation.
2. Fast operation.

Limitations

1. Less accurate.
2. Cannot handle rapid channel variations effectively.

---

2. Closed Loop Power Control

Definition

In closed loop power control:

- Base station continuously monitors signal quality.
- Sends power adjustment commands to mobile users.

Working

```
Mobile Sends Signal
          ↓
Base Station Measures Signal
          ↓
Power Control Command Sent
          ↓
Mobile Adjusts Power
```

Advantages

1. High accuracy.
2. Better interference control.
3. Efficient for dynamic environments.

Limitations

1. More complex.
2. Requires continuous feedback.

Fast Power Control in CDMA

CDMA systems often use:

```
Fast Power Control
```

where power adjustments happen many times per second.

Purpose

- Quickly adapt to changing signal conditions.
- Handle fading and mobility effectively.

Effects of Improper Power Control

Without proper power control:

1. Increased interference.
2. Reduced capacity.
3. Call drops.
4. Poor voice quality.
5. Battery drain.
6. Weak signal detection failure.

Power Control and Interference Relationship

```
High Transmission Power
          ↓
High Interference
          ↓
Reduced Capacity
          ↓
Poor Network Performance
```

Advantages of Efficient Power Control

1. Better communication quality.
2. Increased network capacity.
3. Reduced interference.
4. Improved battery efficiency.
5. Enhanced user experience.

Challenges in Power Control

1. Rapid user mobility.
2. Multipath fading.
3. Delay in feedback signals.
4. Dynamic wireless environment.

Real-Life Example

Suppose multiple users are talking simultaneously in a crowded city:

- Some users are close to the tower.
- Others are far away.

Power control automatically:

- Reduces power of nearby users.
- Increases power of distant users.

This ensures:

- Fair communication for all users.

Comparison with FDMA and TDMA

Feature	FDMA/TDMA	CDMA
Frequency Sharing	Separate channels	Same channel shared
Interference Dependency	Lower	Higher
Need for Power Control	Moderate	Very High

Importance of Power Control in CDMA

Power control is considered the backbone of CDMA because it:

- Maintains signal balance.
- Prevents interference domination.
- Enables efficient frequency reuse.

Without effective power control:

- CDMA systems cannot operate efficiently.

Conclusion

Power control is one of the most important issues in CDMA systems because all users share the same frequency spectrum simultaneously. Without proper power adjustment, strong signals may overpower weak signals, causing the near-far problem and increasing interference. Efficient power control improves signal quality, increases system capacity, conserves battery power, and ensures fair communication among users. Techniques such as open loop and closed loop power control help CDMA systems maintain reliable and efficient wireless communication.

Q3b) Elaborate two basic system architecture of IEEE 802.11 standard? Discuss services offered
by this standard.

Ans 3b) IEEE 802.11 Standard and Its System Architecture

Introduction

IEEE 802.11 is a standard developed by the:

```
Institute of Electrical and Electronics Engineers (IEEE)
```

for Wireless Local Area Networks (WLANs).

It defines:

- Wireless communication protocols
- Network architecture
- Medium access control mechanisms

used in Wi-Fi networks.

The IEEE 802.11 standard allows wireless devices such as:

- Laptops
- Smartphones
- Tablets

to communicate without physical cables.

Features of IEEE 802.11 Standard

1. Wireless communication support.
2. Mobility support.
3. Easy installation.
4. Supports roaming.
5. Flexible networking.

Basic System Architecture of IEEE 802.11

IEEE 802.11 defines two basic system architectures:

1. Infrastructure-Based Architecture
2. Ad-Hoc Architecture

---

1. Infrastructure-Based Architecture

Introduction

Infrastructure mode is the most commonly used architecture in WLANs.

In this architecture:

- Wireless devices communicate through a central device called:

```
Access Point (AP)
```

Components of Infrastructure Architecture

Component	Purpose
Station (STA)	Wireless device
Access Point (AP)	Central communication device
Basic Service Set (BSS)	Group of connected stations
Distribution System (DS)	Connects multiple APs

Working of Infrastructure Mode

1. Wireless stations connect to an Access Point.
2. All communication passes through the AP.
3. AP may connect to wired LAN or Internet.

Diagram of Infrastructure Architecture

```
        Internet
            |
       Access Point
       /    |    \
   STA1  STA2  STA3
```

Basic Service Set (BSS)

Definition

A BSS is a group of stations controlled by a single Access Point.

Extended Service Set (ESS)

Definition

Multiple BSSs connected through a Distribution System form an:

```
Extended Service Set (ESS)
```

Advantages of Infrastructure Mode

1. Centralized management.
2. Better security.
3. Internet connectivity.
4. Supports large networks.
5. Easy roaming support.

Disadvantages

1. Requires Access Point.
2. Higher installation cost.

Applications

- Offices
- Colleges
- Airports
- Homes

---

2. Ad-Hoc Architecture

Introduction

In Ad-Hoc mode:

- Wireless devices communicate directly with each other.
- No Access Point is required.

This architecture is also called:

```
Independent Basic Service Set (IBSS)
```

Components of Ad-Hoc Architecture

Component	Purpose
Wireless Stations	Direct communication

Working of Ad-Hoc Mode

1. Devices connect directly.
2. Each device acts as both:
   - Sender
   - Receiver

Diagram of Ad-Hoc Architecture

```
STA1  ↔  STA2
 ↕        ↕
    STA3
```

Features of Ad-Hoc Architecture

1. No centralized control.
2. Peer-to-peer communication.
3. Temporary network setup.

Advantages of Ad-Hoc Mode

1. Easy setup.
2. No infrastructure required.
3. Low cost.
4. Useful for temporary communication.

Disadvantages

1. Limited range.
2. Lower security.
3. Difficult management.
4. Poor scalability.

Applications

- Emergency communication
- Military communication
- Temporary meetings
- Disaster recovery networks

Comparison Between Infrastructure and Ad-Hoc Architecture

Feature	Infrastructure Mode	Ad-Hoc Mode
Central Device	Access Point required	No Access Point
Communication	Through AP	Direct communication
Management	Centralized	Distributed
Security	Better	Lower
Scalability	High	Limited
Internet Access	Available	Usually unavailable

Services Offered by IEEE 802.11 Standard

IEEE 802.11 provides several services for proper WLAN operation.

These services are divided into two categories:

1. Distribution Services (DS)
2. Station Services (SS)

---

1. Distribution Services (DS)

Distribution services help manage communication between Access Points and stations.

Major distribution services are:

1. Association
2. Reassociation
3. Disassociation
4. Distribution
5. Integration

---

a) Association Service

Definition

Association establishes a connection between:

- Wireless station
  and
- Access Point

Purpose

Allows a device to join the WLAN.

Example

When a smartphone connects to Wi-Fi.

---

b) Reassociation Service

Definition

Reassociation transfers an existing connection from one Access Point to another.

Purpose

Supports:

```
Roaming
```

Example

Moving between Wi-Fi coverage areas in a college campus.

---

c) Disassociation Service

Definition

Terminates the connection between station and Access Point.

Purpose

Releases network resources.

Example

User disconnects from Wi-Fi network.

---

d) Distribution Service

Definition

Transfers data frames between different Access Points.

Purpose

Ensures proper frame routing within WLAN.

---

e) Integration Service

Definition

Connects WLAN to other networks such as:

- Wired LAN
- Internet

Purpose

Enables communication with external networks.

---

2. Station Services (SS)

Station services are directly used by wireless stations.

Major station services are:

1. Authentication
2. Deauthentication
3. Privacy
4. Data Delivery

---

a) Authentication Service

Definition

Verifies the identity of wireless devices before allowing access.

Purpose

Provides network security.

Example

Entering Wi-Fi password before connection.

---

b) Deauthentication Service

Definition

Terminates authentication relationship between station and AP.

Purpose

Removes unauthorized or disconnected users.

---

c) Privacy Service

Definition

Provides data security using encryption techniques.

Purpose

Protects wireless communication from unauthorized access.

Example

WPA2 and WPA3 encryption.

---

d) Data Delivery Service

Definition

Responsible for transmitting data packets between devices.

Summary of IEEE 802.11 Services

Service	Purpose
Association	Connect station to AP
Reassociation	Support roaming
Disassociation	Disconnect station
Authentication	Verify identity
Privacy	Secure communication
Distribution	Frame forwarding
Integration	Connect external networks

Advantages of IEEE 802.11 Standard

1. Wireless mobility.
2. Easy installation.
3. Cost-effective networking.
4. Supports roaming.
5. Flexible communication.

Disadvantages of IEEE 802.11

1. Security vulnerabilities.
2. Signal interference.
3. Limited range.
4. Lower speed compared to wired LAN.

Applications of IEEE 802.11

Application Area	Example
Home Networks	Wi-Fi routers
Offices	Wireless LAN
Airports	Public Wi-Fi
Educational Institutions	Campus networking

Conclusion

IEEE 802.11 is the standard used for wireless LAN communication and forms the basis of modern Wi-Fi networks. It supports two basic architectures: Infrastructure-Based Architecture and Ad-Hoc Architecture. Infrastructure mode uses Access Points for centralized communication, while Ad-Hoc mode allows direct peer-to-peer communication. IEEE 802.11 also provides important services such as association, authentication, reassociation, privacy, and data delivery to ensure efficient and secure wireless networking. These features make IEEE 802.11 an essential technology in modern mobile computing and wireless communication systems.

Q3c) Compare hybrid channel allocation strategy with the static channel allocation strategies?

Ans 3c) Comparison Between Hybrid Channel Allocation and Static Channel Allocation Strategies

Introduction

In cellular mobile communication systems, available radio channels must be allocated efficiently among different cells to support communication services such as:

- Voice calls
- Messaging
- Mobile internet

The process of assigning channels to cells is called:

```
Channel Allocation
```

Efficient channel allocation is important for:

- Reducing interference
- Increasing network capacity
- Improving Quality of Service (QoS)

Two important channel allocation strategies are:

1. Static Channel Allocation (SCA)
2. Hybrid Channel Allocation (HCA)

Channel Allocation

Definition

Channel allocation is the process of distributing available radio frequency channels among cells in a cellular network.

Objectives of Channel Allocation

1. Efficient spectrum utilization.
2. Reduced interference.
3. Increased capacity.
4. Improved communication quality.
5. Reduced call blocking.

Static Channel Allocation (SCA)

Definition

In Static Channel Allocation, a fixed number of channels are permanently assigned to each cell.

These channels cannot be shared dynamically with other cells.

Working of Static Channel Allocation

1. Total channels are divided among cells.
2. Each cell receives predefined channels.
3. Calls use only allocated channels.

Example

Suppose:

- Total channels = 100
- 10 cells in network

Then:

- Each cell may receive 10 fixed channels.

Diagram of SCA

```
Cell A → Fixed Channels
Cell B → Fixed Channels
Cell C → Fixed Channels
```

Characteristics of SCA

1. Fixed channel assignment.
2. Simple implementation.
3. No dynamic borrowing.
4. Predictable operation.

Advantages of Static Channel Allocation

1. Simple network management.
2. Easy implementation.
3. Low computational complexity.
4. Stable channel planning.

Disadvantages of Static Channel Allocation

1. Poor spectrum utilization.
2. High call blocking during heavy traffic.
3. Unused channels remain idle.
4. Cannot adapt to traffic variation.

Hybrid Channel Allocation (HCA)

Definition

Hybrid Channel Allocation combines the features of:

- Static Channel Allocation
  and
- Dynamic Channel Allocation

Some channels are assigned permanently, while others are allocated dynamically based on traffic demand.

Working of Hybrid Channel Allocation

1. Each cell receives fixed channels initially.
2. Additional channels are dynamically assigned when traffic increases.
3. Channels can be borrowed from a common pool.

Example

Suppose:

- A cell has 5 fixed channels.
- During heavy traffic, extra channels are allocated dynamically.

Diagram of HCA

```
Fixed Channels
       +
Dynamic Shared Channels
```

Characteristics of Hybrid Channel Allocation

1. Combination of static and dynamic allocation.
2. Adaptive to traffic conditions.
3. Better channel utilization.
4. Flexible operation.

Advantages of Hybrid Channel Allocation

1. Better spectrum efficiency.
2. Reduced call blocking.
3. Improved network capacity.
4. Handles traffic variations effectively.
5. Flexible resource allocation.

Disadvantages of Hybrid Channel Allocation

1. More complex implementation.
2. Higher computational overhead.
3. Complex channel management.
4. Requires dynamic monitoring.

Comparison Between Hybrid and Static Channel Allocation

Feature	Static Channel Allocation (SCA)	Hybrid Channel Allocation (HCA)
Channel Assignment	Fixed	Fixed + Dynamic
Flexibility	Low	High
Traffic Adaptation	Poor	Good
Spectrum Utilization	Less efficient	More efficient
Complexity	Simple	Complex
Call Blocking Probability	Higher	Lower
Resource Utilization	Limited	Better
Computational Requirement	Low	Higher

Channel Utilization Comparison

Static Allocation

```
Unused channels may remain idle
even if nearby cells need channels.
```

Hybrid Allocation

```
Extra channels can be dynamically
allocated where traffic is high.
```

Traffic Handling Comparison

Situation	SCA Performance	HCA Performance
Low Traffic	Good	Good
Heavy Traffic	Poor	Better
Uneven Traffic Distribution	Inefficient	Efficient

Spectrum Efficiency

Static Allocation

- Fixed channels may remain unused.
- Wastage of bandwidth possible.

Hybrid Allocation

- Dynamic sharing improves spectrum usage.
- Channels are allocated where needed.

Call Blocking Probability

Static Allocation

When all fixed channels are busy:

- New calls are blocked.

Hybrid Allocation

Dynamic channels reduce:

```
Call Blocking
```

during peak traffic.

Network Capacity Comparison

Allocation Method	Capacity
Static Allocation	Lower
Hybrid Allocation	Higher

Applications of Static Channel Allocation

1. Small cellular systems.
2. Low traffic networks.
3. Simple communication systems.

Applications of Hybrid Channel Allocation

1. Modern mobile networks.
2. High traffic urban areas.
3. Cellular systems with variable traffic.

Real-Life Example

Static Allocation Example

In a rural area:

- Traffic is stable and predictable.
- Fixed channels may be sufficient.

Hybrid Allocation Example

In a city:

- Traffic changes continuously.
- Hybrid allocation dynamically manages channels efficiently.

Why Hybrid Allocation is Preferred in Modern Networks

Modern mobile networks experience:

- Variable traffic
- Sudden congestion
- Mobility of users

Hybrid allocation provides:

1. Better flexibility.
2. Improved QoS.
3. Higher efficiency.

Challenges in Hybrid Channel Allocation

1. Complex algorithms.
2. Dynamic monitoring requirement.
3. Increased signaling overhead.

Advantages of Channel Allocation Strategies

Strategy	Main Advantage
Static Allocation	Simplicity
Hybrid Allocation	Efficiency and flexibility

Conclusion

Static Channel Allocation assigns fixed channels permanently to cells, making it simple but less efficient in handling changing traffic conditions. Hybrid Channel Allocation combines both static and dynamic approaches, allowing better spectrum utilization and reduced call blocking. Although hybrid allocation is more complex, it provides greater flexibility, improved capacity, and efficient handling of variable traffic patterns. Therefore, modern cellular networks generally prefer hybrid channel allocation strategies for efficient mobile communication management.

Q4a)

a. Identify various issues of data management in mobile computing? Describe data replication for
mobile computer. Explain working principles of peer-to-peer reconciliation and Rumor replicated file system.

Ans 4a) Data Management in Mobile Computing

Introduction

Mobile computing allows users to access and exchange data while moving through different locations using wireless communication networks. Mobile devices such as:

- Smartphones
- Tablets
- Laptops

continuously interact with distributed databases and remote servers.

Managing data efficiently in such environments is difficult because mobile systems face problems such as:

- Frequent disconnections
- Limited bandwidth
- Device mobility
- Battery constraints

Therefore, effective:

```
Data Management
```

is essential in mobile computing systems.

Data Management in Mobile Computing

Definition

Data management in mobile computing refers to the techniques and processes used for:

- Storing
- Accessing
- Updating
- Synchronizing

data efficiently in mobile and wireless environments.

Issues of Data Management in Mobile Computing

Mobile computing environments create several challenges for data management.

---

1. Limited Bandwidth

Problem

Wireless networks provide lower bandwidth compared to wired networks.

Effects

1. Slow data transfer.
2. Increased transmission delay.
3. Reduced communication efficiency.

Example

Large file downloads become slower on mobile networks.

---

2. Frequent Disconnection

Problem

Mobile devices may lose connectivity because of:

- Weak signals
- User mobility
- Network failures

Effects

1. Interrupted communication.
2. Incomplete transactions.
3. Data inconsistency.

Example

Internet disconnection during online payment.

---

3. Mobility of Users

Problem

Users continuously move between network locations.

Effects

1. Dynamic address changes.
2. Difficult session management.
3. Increased handoff overhead.

---

4. Limited Battery Power

Problem

Mobile devices operate using battery power.

Effects

1. Restricted processing.
2. Limited communication time.
3. Energy-efficient protocols required.

---

5. Data Consistency

Problem

Copies of data stored at multiple locations may become inconsistent.

Effects

1. Incorrect information.
2. Synchronization conflicts.

Example

Different versions of same file on two devices.

---

6. Security and Privacy

Problem

Wireless communication is vulnerable to:

- Hacking
- Eavesdropping
- Unauthorized access

Effects

1. Data leakage.
2. Security threats.

---

7. Limited Storage Capacity

Problem

Mobile devices have limited memory compared to servers.

Effects

1. Restricted local storage.
2. Need for data compression.

---

8. Scalability Issues

Problem

Large numbers of mobile users increase network load.

Effects

1. Congestion.
2. Reduced performance.

Summary of Data Management Issues

Issue	Effect
Limited Bandwidth	Slow communication
Frequent Disconnection	Interrupted services
Mobility	Dynamic network changes
Battery Constraints	Limited processing
Data Consistency	Synchronization problems

Issue	Effect
Security	Unauthorized access
Limited Storage	Reduced local capacity

Data Replication in Mobile Computing

Introduction

To improve data availability and performance, mobile systems often use:

```
Data Replication
```

Definition of Data Replication

Data replication is the process of creating and maintaining multiple copies of the same data at different locations in a network.

Purpose of Data Replication

1. Improve data availability.
2. Reduce access delay.
3. Support disconnected operation.
4. Increase fault tolerance.
5. Improve system reliability.

Working of Data Replication

```
Original Data
        ↓
Copies Created
        ↓
Stored at Multiple Locations
        ↓
Synchronization Performed
```

Types of Replication

Type	Description
Full Replication	Entire database copied
Partial Replication	Selected data copied

Advantages of Data Replication

1. Faster data access.
2. Better reliability.
3. Offline data availability.
4. Reduced server load.

Disadvantages of Data Replication

1. Synchronization complexity.
2. Increased storage requirement.
3. Data inconsistency risk.

Example of Data Replication

Cloud storage applications such as:

- Google Drive
- Dropbox

store copies of files on multiple servers.

Peer-to-Peer Reconciliation

Introduction

In mobile computing, replicated data copies may become inconsistent due to:

- Offline operation
- Independent updates

Peer-to-peer reconciliation helps synchronize these copies.

Definition

Peer-to-peer reconciliation is a synchronization technique in which mobile devices directly exchange updates with each other to maintain data consistency.

Working Principle of Peer-to-Peer Reconciliation

1. Each device maintains its own data copy.
2. Devices may update data independently.
3. When devices reconnect:
    - Updates are exchanged.
4. Conflicts are detected and resolved.

Workflow

```
Device A Updates Data
          ↓
Device B Updates Data
          ↓
Devices Reconnect
          ↓
Update Exchange Occurs
          ↓
Conflicts Resolved
```

Conflict Resolution

Conflicts occur when:

- Same data item is modified differently.

Resolution methods include:

1. Latest update wins.
2. User intervention.
3. Version comparison.

Advantages of Peer-to-Peer Reconciliation

1. Supports disconnected operation.
2. Reduces central server dependency.
3. Efficient synchronization.

Disadvantages

1. Conflict management complexity.
2. Increased synchronization overhead.

Example

Two employees edit same document offline on separate devices and synchronize changes later.

Rumor Replicated File System

Introduction

Rumor Replicated File System is a distributed file replication mechanism used in mobile and distributed systems.

It spreads updates gradually among nodes like:

```
Rumors
```

in social communication.

Definition

Rumor Replicated File System is a replication technique in which updates are propagated incrementally among distributed nodes through periodic communication.

Working Principle of Rumor Replicated File System

1. Each node stores replicated file copies.
2. When a file changes:
    - Update information is generated.
3. Nodes periodically exchange update information.
4. Updates gradually spread across the network.

Workflow of Rumor Replication

```
File Updated at Node A
          ↓
Node A Shares Update with Node B
          ↓
Node B Shares Update with Node C
          ↓
Update Gradually Spreads
```

Characteristics of Rumor Replication

1. Distributed synchronization.
2. Incremental update spreading.
3. Eventual consistency.

Advantages of Rumor Replicated File System

1. Scalable synchronization.
2. Reduced communication overhead.
3. Suitable for mobile environments.
4. Fault tolerant.

Disadvantages

1. Delayed consistency.
2. Temporary data mismatch possible.
3. Complex conflict handling.

Example

Mobile collaborative applications where users synchronize shared files gradually over wireless networks.

Comparison Between Peer-to-Peer Reconciliation and Rumor Replication

Feature	Peer-to-Peer Reconciliation	Rumor Replicated File System
Communication	Direct synchronization	Gradual update propagation
Consistency	Faster consistency	Eventual consistency
Complexity	Moderate	Higher
Scalability	Moderate	High
Central Server	Not required	Not required

Applications of Data Replication in Mobile Computing

Application	Usage
Cloud Storage	File synchronization
Mobile Banking	Data backup
Collaborative Apps	Shared editing
Social Media Apps	Offline synchronization

Importance of Data Replication

1. Improves reliability.
2. Enables offline access.
3. Reduces access delay.
4. Supports mobility.

Challenges in Replication Systems

1. Synchronization conflicts.
2. Increased storage cost.
3. Network overhead.
4. Security risks.

Conclusion

Data management in mobile computing faces several challenges such as limited bandwidth, mobility, disconnections, security concerns, and data consistency issues. Data replication is an important technique used to improve availability, reliability, and performance by maintaining multiple copies of data. Peer-to-peer reconciliation allows direct synchronization between devices, while the Rumor Replicated File System spreads updates gradually across distributed nodes. These techniques help mobile computing systems maintain efficient and reliable data access in dynamic wireless environments.

Q4b) Discuss life cycle of mobile agent based computing with suitable diagram

Ans 4b) Life Cycle of Mobile Agent Based Computing

Introduction

Mobile Agent technology is an important concept in mobile computing and distributed systems. A mobile agent is a software program that can move from one computer or network node to another while carrying:

- Code

- Data
- Execution state

Mobile agents perform tasks autonomously and continue execution at remote locations.

This technology is used in:

- Distributed computing
- Network management
- E-commerce
- Information retrieval
- Mobile communication systems

The operation of a mobile agent follows a sequence of stages known as the:

```
Life Cycle of Mobile Agent
```

Mobile Agent

Definition

A Mobile Agent is a software entity that can migrate autonomously across network nodes while performing tasks on behalf of a user or application.

Features of Mobile Agents

1. Mobility
2. Autonomy
3. Communication capability
4. Adaptability
5. Persistence

Need for Mobile Agents

Mobile agents are used because they:

1. Reduce network traffic.
2. Support disconnected operations.
3. Improve distributed processing.
4. Reduce communication delay.
5. Support asynchronous execution.

Life Cycle of Mobile Agent

Introduction

The life cycle of a mobile agent represents different states through which the agent passes during execution and migration.

The main stages are:

1. Creation
2. Dispatch / Migration
3. Execution
4. Suspension
5. Resumption
6. Termination

Diagram of Mobile Agent Life Cycle

```
   Creation
       ↓
Dispatch/Migration
       ↓
   Execution
       ↓
Suspension State
       ↓
   Resumption
       ↓
   Termination
```

1. Creation State

Definition

The creation state is the initial stage where the mobile agent is generated and initialized.

Activities Performed

1. Agent code creation.
2. Memory allocation.
3. Initialization of variables.
4. Setting execution goals.

Working

- The user or application creates the mobile agent.
- Initial execution environment is prepared.

Example

A mobile shopping assistant agent is created to search product prices on multiple servers.

Importance

This stage defines:

- Agent identity
- Tasks
- Execution parameters

---

2. Dispatch / Migration State

Definition

In this stage, the mobile agent moves from one host or network node to another.

This process is called:

```
Migration
```

Activities Performed

1. Agent packaging.
2. State saving.
3. Network transfer.
4. Arrival at destination host.

Types of Migration

Type	Description
Strong Migration	Complete execution state transferred
Weak Migration	Only code and data transferred

Example

The shopping agent moves from:

- User device
  to
- E-commerce server

Advantages

1. Reduces network communication.
2. Supports distributed processing.

---

3. Execution State

Definition

In this state, the mobile agent executes tasks at the destination host.

Activities Performed

1. Data collection.
2. Resource access.

3. Information processing.
4. Communication with local resources.

Example

The shopping agent compares product prices on the server.

Characteristics

1. Autonomous execution.
2. Local processing.
3. Efficient resource utilization.

Importance

Execution near data source reduces:

- Communication overhead
- Network traffic

---

4. Suspension State

Definition

Suspension is the temporary stopping of agent execution.

Causes of Suspension

1. Network failure.
2. Resource unavailability.
3. Waiting for external events.
4. Low battery conditions.

Activities Performed

1. Execution state saved.
2. Agent paused temporarily.

Example

Agent waits for server response before continuing.

Importance

Suspension supports:

- Fault tolerance
- Resource management

---

5. Resumption State

Definition

In the resumption state, the suspended mobile agent restarts execution from the saved state.

Activities Performed

1. Restore execution state.
2. Continue pending tasks.

Example

After network reconnection:

- Agent resumes data collection.

Importance

Ensures:

- Continuity of operations
- Reliable execution

6. Termination State

Definition

Termination is the final stage where the mobile agent completes its task and stops execution.

Activities Performed

1. Return results.
2. Release resources.
3. Delete execution state.

Example

Shopping agent sends product comparison results to the user and terminates.

Importance

1. Prevents unnecessary resource usage.
2. Frees system memory.

Complete Working of Mobile Agent Life Cycle

```
User Creates Agent
          ↓
Agent Migrates to Remote Host
          ↓
Agent Executes Task
          ↓
Agent Suspended if Needed
          ↓
Agent Resumes Execution
          ↓
Agent Completes Task
          ↓
Agent Terminates
```

States of Mobile Agent in Summary

State	Purpose
Creation	Initialize agent
Migration	Move to another host
Execution	Perform tasks
Suspension	Pause execution
Resumption	Continue execution
Termination	End execution

Advantages of Mobile Agent Based Computing

1. Reduced network traffic.
2. Faster execution.
3. Supports disconnected operations.
4. Better scalability.
5. Reduced communication latency.

Disadvantages of Mobile Agent Based Computing

1. Security risks.
2. Complex implementation.
3. Resource management difficulty.
4. Agent authentication challenges.

Applications of Mobile Agents

Application	Usage
E-Commerce	Product search
Network Management	Monitoring systems
Information Retrieval	Distributed search
Mobile Banking	Secure transactions
Distributed Computing	Task execution

Real-Life Example

Suppose a user wants flight ticket information from multiple airline servers.

Instead of repeatedly sending requests:

- A mobile agent travels to airline servers,
- Collects data locally,
- Returns optimized results to the user.

This reduces:

- Network load
- Communication delay

Importance of Mobile Agent Life Cycle

Understanding the life cycle helps in:

1. Efficient agent management.

2. Reliable execution.
3. Better resource allocation.
4. Improved fault tolerance.

Challenges in Mobile Agent Systems

1. Security threats.
2. Agent synchronization.
3. Host compatibility.
4. Fault handling.
5. Migration overhead.

Conclusion

The life cycle of mobile agent based computing consists of several stages including creation, migration, execution, suspension, resumption, and termination. These stages allow mobile agents to move across distributed systems and perform tasks autonomously. Mobile agents reduce network traffic, improve distributed processing efficiency, and support mobile computing environments effectively. Understanding the mobile agent life cycle is essential for designing secure and efficient distributed mobile systems.

Q4c) Discuss design Goals and Architecture of CODA file system. What is side effect in Coda file
System

Ans 4c) CODA File System: Design Goals, Architecture and Side Effects

Introduction

The CODA File System is a distributed file system developed at:

```
Carnegie Mellon University
```

It is designed mainly for:

- Mobile computing
- Distributed environments
- Fault-tolerant file access

CODA is an advanced version of the:

```
Andrew File System (AFS)
```

It supports:

- Disconnected operation
- Data replication
- High availability
- Mobile user support

CODA is widely known for providing reliable file access even when network connectivity is unstable.

CODA File System

Definition

CODA is a distributed file system designed to provide highly available and efficient file access in mobile and distributed computing environments.

Features of CODA File System

1. Disconnected operation.
2. File replication.
3. High availability.
4. Fault tolerance.
5. Client-side caching.
6. Mobile computing support.

Design Goals of CODA File System

The CODA file system was designed with several important goals.

---

1. High Availability

Goal

Files should remain accessible even if:

- Some servers fail
  or
- Network disconnections occur.

Importance

Ensures uninterrupted file access.

Example

A mobile user can continue reading cached files even during network failure.

---

2. Support for Disconnected Operation

Goal

Mobile users should continue working even when disconnected from the network.

Working

Files are stored temporarily in local cache.

Importance

Very useful for:

- Mobile users
- Wireless communication environments

---

3. Scalability

Goal

The system should efficiently support:

- Large numbers of users
- Large distributed networks

Importance

Improves system performance in large environments.

---

4. Fault Tolerance

Goal

System should continue functioning despite:

- Server crashes
- Communication failures

Importance

Improves reliability and robustness.

---

5. Security

Goal

Protect data from:

- Unauthorized access
- Security attacks

Techniques Used

1. Authentication
2. Access control
3. Encryption

---

6. Transparency

Goal

Users should access files without worrying about:

- File locations
- Replication details

Importance

Simplifies user interaction.

---

7. Efficient Performance

Goal

Reduce:

- Communication delay
- Network traffic

Techniques Used

1. Client caching
2. Replication
3. Local processing

Summary of Design Goals

Design Goal	Purpose
High Availability	Continuous file access
Disconnected Operation	Mobile user support
Scalability	Large network support
Fault Tolerance	Reliable operation
Security	Data protection
Transparency	Easy file access
Performance	Faster operations

Architecture of CODA File System

Introduction

CODA follows a:

```
Client-Server Architecture
```

It consists of:

- Clients
- Servers
- Local cache
- Volume Storage Group (VSG)

Components of CODA Architecture

1. Client
2. Server
3. Venus
4. Vice
5. Volume Storage Group (VSG)
6. Cache Manager

Diagram of CODA Architecture

```
     CODA Servers
  (Vice Servers)
    /    |    \
   /     |     \
----------------
      Network
----------------
         |
   Mobile Client
      (Venus)
         |
   Local Cache
```

1. Client

Definition

The client is the mobile user system that accesses files from CODA servers.

Functions

1. Request file access.
2. Store cached files.
3. Support disconnected operation.

Example

Laptop or mobile workstation.

2. Server (Vice)

Definition

CODA servers, called:

`Vice Servers`

store the actual files and manage replication.

Functions

1. File storage.
2. Replication management.
3. Access control.
4. Synchronization.

Importance

Ensures data availability and consistency.

---

3. Venus (Client Cache Manager)

Definition

Venus is the client-side cache manager in CODA.

It acts as an interface between:

- Client applications
  and
- Vice servers

Functions of Venus

1. File caching.
2. Disconnected operation support.
3. Synchronization.
4. Communication with servers.

Importance

Improves performance by reducing network access.

---

4. Volume Storage Group (VSG)

Definition

A VSG is a group of servers that maintain replicated copies of data volumes.

Purpose

Provides:

- High availability
- Fault tolerance

Working

If one server fails:

- Another replica server provides data.

---

5. Cache

Definition

Local storage area on the client for temporarily storing files.

Functions

1. Offline file access.

2. Faster data retrieval.
3. Reduced network traffic.

Working of CODA File System

Step 1: File Request

Client requests a file.

Step 2: Venus Checks Cache

- If file exists locally:
    - File is accessed directly.
- Otherwise:
    - Request sent to server.

Step 3: File Retrieval

Vice server sends file to client.

Step 4: Local Caching

Venus stores file in local cache.

Step 5: Disconnected Operation

User can continue accessing cached files even without network connection.

Step 6: Reconnection and Synchronization

When network reconnects:

- Changes are synchronized with servers.

Advantages of CODA File System

1. Supports mobile computing.
2. Provides disconnected operation.
3. High fault tolerance.
4. Improved performance through caching.
5. Better availability using replication.

Disadvantages of CODA File System

1. Complex synchronization.
2. Cache consistency challenges.
3. Increased storage overhead.

Side Effects in CODA File System

Definition

Side effects in CODA file system refer to inconsistencies or conflicts that occur when multiple disconnected clients independently modify replicated files and later reconnect to synchronize changes.

Cause of Side Effects

Side effects mainly occur because of:

1. Disconnected operation.
2. Concurrent updates.
3. Replicated file modifications.

Example of Side Effect

Suppose:

- User A edits a file offline.
- User B edits the same file offline.

When both reconnect:

- Different versions of the same file exist.

This creates:

```
Conflict or Side Effect
```

Types of Side Effects

Type	Description
Update Conflict	Multiple conflicting updates
Data Inconsistency	Different file versions
Synchronization Conflict	Merge problems

Consequences of Side Effects

1. Data inconsistency.
2. Synchronization difficulty.
3. File conflicts.
4. Possible data loss.

Handling Side Effects in CODA

CODA uses:

```
Conflict Resolution Mechanisms
```

to manage side effects.

Conflict Resolution Methods

1. Version comparison.
2. User-assisted conflict resolution.
3. Automatic merge techniques.

Importance of Side Effect Handling

Proper handling ensures:

1. Data consistency.
2. Reliable synchronization.
3. Correct file versions.

Real-Life Example

Suppose two employees edit the same office document while disconnected from the network.

After reconnection:

- CODA detects conflicting updates.
- Conflict resolution process merges or selects correct version.

Comparison Between AFS and CODA

Feature	AFS	CODA
Disconnected Operation	Limited	Supported
Replication	Basic	Advanced
Mobile Support	Less	Better
Fault Tolerance	Moderate	High

Applications of CODA File System

Application Area	Usage

Application Area	Usage
Mobile Computing	Offline file access
Distributed Systems	Shared file management
Enterprise Networks	Reliable file sharing

Conclusion

The CODA file system is a distributed file system specially designed for mobile and distributed computing environments. Its design goals include high availability, disconnected operation, fault tolerance, scalability, and security. CODA uses a client-server architecture with components such as Venus, Vice servers, and replicated storage groups to provide reliable file access. Side effects in CODA occur because of conflicting updates during disconnected operation, and these conflicts are managed through synchronization and reconciliation techniques. CODA remains an important example of distributed file system design for mobile computing systems.

Q5a) Discuss characteristics of Mobile Adhoc Networks? Why is it not possible to use circuit switching in Adhoc networks?

Ans 5a) Mobile Ad Hoc Networks (MANETs)

Introduction

A Mobile Ad Hoc Network (MANET) is a self-configuring wireless network formed by mobile devices without using any fixed infrastructure such as:

- Base stations
- Routers
- Access points

In MANET:

- Mobile nodes communicate directly with each other.
- Nodes can also act as routers for forwarding data.

These networks are highly useful in environments where fixed infrastructure is unavailable or difficult to establish.

Definition of MANET

A Mobile Ad Hoc Network (MANET) is a collection of wireless mobile nodes that dynamically form a temporary network without centralized administration or fixed infrastructure.

Features of MANET

1. Infrastructure-less network.
2. Dynamic topology.
3. Wireless communication.
4. Multi-hop routing.
5. Distributed operation.

Characteristics of Mobile Ad Hoc Networks

MANETs possess several unique characteristics.

---

1. Infrastructure-less Network

Explanation

MANET does not require:

- Base stations
- Central controllers
- Fixed routers

Nodes communicate directly.

Importance

1. Easy deployment.
2. Suitable for temporary networks.

Example

Communication during natural disasters where infrastructure is damaged.

---

2. Dynamic Topology

Explanation

Mobile nodes continuously move, causing frequent changes in network topology.

Effects

1. Routes change frequently.
2. Connections may break dynamically.

Example

Vehicles moving in a VANET (Vehicular Ad Hoc Network).

---

3. Distributed Operation

Explanation

There is no centralized control in MANET.

Every node:

- Participates in routing
- Shares network responsibilities

Importance

Improves flexibility and fault tolerance.

---

4. Multi-hop Communication

Explanation

If two nodes are outside direct communication range:

- Intermediate nodes forward packets.

Example

```
Node A → Node B → Node C
```

Node B acts as router between A and C.

Importance

Extends communication range.

---

5. Mobility of Nodes

Explanation

Nodes can move freely in any direction.

Effects

1. Frequent route updates.
2. Dynamic connectivity.

Example

Soldiers carrying mobile communication devices in battlefield networks.

---

6. Limited Bandwidth

Explanation

Wireless channels provide lower bandwidth than wired networks.

Effects

1. Reduced data transfer speed.
2. Network congestion possibility.

---

7. Energy Constraints

Explanation

Mobile nodes operate on battery power.

Effects

1. Limited operating time.
2. Need for energy-efficient protocols.

Example

Battery-powered smartphones in MANET.

---

8. Limited Physical Security

Explanation

Wireless communication is vulnerable to:

- Eavesdropping
- Hacking
- Unauthorized access

Effects

1. Security threats.
2. Data interception risk.

---

9. Self-Configuration

Explanation

Nodes automatically configure themselves without manual setup.

Importance

1. Easy deployment.
2. Fast network formation.

---

10. Frequent Route Changes

Explanation

Due to node mobility:

- Communication routes change frequently.

Effects

1. Routing complexity increases.
2. Packet delivery challenges occur.

Summary of Characteristics

Characteristic	Description
Infrastructure-less	No fixed infrastructure
Dynamic Topology	Frequent topology changes
Distributed Operation	No centralized control
Multi-hop Routing	Intermediate nodes forward packets
Mobility	Nodes move freely
Limited Bandwidth	Lower wireless capacity
Energy Constraints	Battery-powered devices
Self-Configuration	Automatic setup

Advantages of MANET

1. Easy deployment.
2. No infrastructure cost.
3. Flexible communication.
4. Useful in remote areas.
5. Supports mobility.

Disadvantages of MANET

1. Security vulnerabilities.
2. Limited battery life.
3. Dynamic routing complexity.
4. Lower bandwidth.
5. Frequent disconnections.

Applications of MANET

Application	Usage
Military Communication	Battlefield networks
Disaster Recovery	Emergency communication
Vehicular Networks	Vehicle communication
Sensor Networks	Monitoring systems
Temporary Meetings	Conference networking

Why Circuit Switching Cannot Be Used in Ad Hoc Networks

Introduction

Circuit switching is a communication technique used in traditional telephone networks where:

- A dedicated communication path is established between sender and receiver before data transfer begins.

In MANETs, using circuit switching is not practical because of the unique characteristics of ad hoc networks.

Circuit Switching

Definition

Circuit switching is a communication method in which a dedicated communication channel is established for the entire duration of communication.

Example

Traditional telephone systems use circuit switching.

Working of Circuit Switching

```
Path Established
        ↓
Dedicated Connection Reserved
        ↓
Communication Performed
        ↓
Connection Released
```

Reasons Why Circuit Switching is Not Suitable for MANET

---

1. Dynamic Topology

Explanation

In MANET:

- Nodes move continuously.
- Network paths change frequently.

Circuit switching requires:

- Stable dedicated paths.

Problem

Dedicated circuits break frequently due to mobility.

Example

If an intermediate node moves away:

- Entire communication path fails.

---

2. Frequent Route Changes

Explanation

MANET routes are highly dynamic.

Circuit switching cannot efficiently handle:

- Continuous path changes.

Result

Frequent circuit re-establishment would be required.

---

3. Limited Bandwidth

Explanation

Wireless bandwidth is limited in MANET.

Circuit switching reserves bandwidth even when:

- No data is being transmitted.

Problem

Bandwidth wastage occurs.

Packet Switching Advantage

Packet switching uses bandwidth only when needed.

---

4. No Fixed Infrastructure

Explanation

Circuit switching requires:

- Centralized switching systems
- Fixed routing infrastructure

MANET lacks:

- Base stations
- Central controllers

Problem

Dedicated circuit management becomes impossible.

---

5. High Mobility

Explanation

Nodes frequently enter and leave the network.

Circuit switching assumes:

- Stable communication endpoints.

Problem

Maintaining fixed circuits becomes impractical.

---

6. Energy Constraints

Explanation

Circuit switching requires continuous connection maintenance.

Problem

Continuous circuits consume:

- More battery power
- More network resources

---

7. Inefficient Resource Utilization

Explanation

In circuit switching:

- Resources remain reserved even during idle periods.

Problem

Wireless resources are wasted in MANET environments.

---

8. Scalability Issues

Explanation

Large MANETs have:

- Many mobile nodes
- Frequent topology changes

Problem

Managing dedicated circuits becomes highly complex.

Packet Switching vs Circuit Switching in MANET

Feature	Circuit Switching	Packet Switching
Connection Type	Dedicated path	Dynamic packets
Resource Usage	Reserved continuously	Used when needed
Suitability for Mobility	Poor	Good
Topology Adaptation	Difficult	Easy
Bandwidth Efficiency	Low	High
Scalability	Limited	Better

Why Packet Switching is Preferred in MANET

MANETs generally use:

```
Packet Switching
```

because:

1. It handles mobility efficiently.
2. No dedicated path required.
3. Better bandwidth utilization.
4. Supports dynamic routing.

Real-Life Example

Suppose soldiers communicate using MANET during military operations:

- Soldiers continuously move.
- Communication paths change dynamically.

Circuit switching would fail because:

- Dedicated routes cannot remain stable.

Packet switching works better because:

- Packets can dynamically choose available routes.

Conclusion

Mobile Ad Hoc Networks (MANETs) are self-configuring wireless networks characterized by dynamic topology, distributed operation, mobility, multi-hop communication, and lack of fixed infrastructure. These features make MANETs highly flexible and suitable for mobile and temporary communication environments. Circuit switching is not suitable for MANETs because it requires stable dedicated communication paths, centralized infrastructure, and continuous resource reservation. Due to dynamic topology and node mobility, packet switching is preferred in MANETs as it provides efficient, flexible, and adaptive communication.

Q5b) List out the problems in message routing in Adhoc wireless networks? What are advantages
and disadvantage of reactive and proactive protocols?

Ans 5b) Problems in Message Routing in Ad Hoc Wireless Networks

Introduction

In Ad Hoc Wireless Networks or:

```
MANETs (Mobile Ad Hoc Networks)
```

mobile nodes communicate without fixed infrastructure such as:

- Base stations
- Routers
- Access points

Each node acts as:

- Sender
- Receiver
- Router

Because of node mobility and dynamic topology, message routing in ad hoc networks becomes very challenging.

Routing protocols must continuously discover and maintain routes between mobile nodes.

Message Routing in Ad Hoc Networks

Definition

Message routing is the process of selecting and maintaining paths for transmitting data packets from source to destination in an ad hoc wireless network.

Problems in Message Routing in Ad Hoc Wireless Networks

Ad hoc networks face several routing challenges.

---

1. Dynamic Topology

Explanation

Nodes move frequently, causing:

- Continuous topology changes
- Frequent route breakage

Effects

1. Routing tables become outdated.
2. Frequent route rediscovery required.

Example

Vehicles moving in VANET networks.

---

2. Limited Bandwidth

Explanation

Wireless communication channels have limited bandwidth.

Effects

1. Congestion occurs.
2. Routing overhead increases.
3. Data transmission becomes slower.

---

3. Frequent Link Failures

Explanation

Wireless links may fail due to:

- Node movement
- Signal fading
- Interference

Effects

1. Packet loss.
2. Communication interruption.
3. Route instability.

---

4. Limited Battery Power

Explanation

Mobile nodes operate on batteries.

Effects

1. Nodes may shut down unexpectedly.
2. Routing paths become unavailable.

Importance

Routing protocols must be:

```
Energy Efficient
```

---

5. Scalability Issues

Explanation

As network size increases:

- Routing becomes more complex.

Effects

1. Large routing tables.
2. Increased control overhead.
3. Reduced performance.

---

6. Security Threats

Explanation

Wireless communication is vulnerable to:

- Eavesdropping
- Spoofing
- Denial of Service attacks

Effects

1. Unauthorized access.
2. Routing manipulation.

Example

Malicious nodes may send fake routing information.

---

7. Hidden Terminal Problem

Explanation

Two nodes outside each other's range may transmit simultaneously to the same node.

Effects

1. Packet collision.
2. Data corruption.

Example

```
Node A → Node B ← Node C
```

A and C cannot hear each other.

---

8. Routing Overhead

Explanation

Frequent route updates generate:

- Extra control packets

Effects

1. Increased bandwidth consumption.
2. Reduced data transmission efficiency.

---

9. Quality of Service (QoS) Issues

Explanation

Maintaining:

- Delay
- Throughput
- Reliability

is difficult in MANET.

Effects

Poor multimedia communication quality.

---

10. Multi-hop Communication Complexity

Explanation

Data often passes through multiple intermediate nodes.

Effects

1. Increased delay.
2. Higher probability of route failure.

Summary of Routing Problems

Problem	Effect
Dynamic Topology	Frequent route changes
Limited Bandwidth	Congestion
Link Failure	Communication interruption
Battery Constraints	Node unavailability
Security Threats	Data attacks
Routing Overhead	Reduced efficiency
Scalability	Complex routing

Routing Protocols in Ad Hoc Networks

Routing protocols in MANET are mainly divided into:

1. Reactive Routing Protocols
2. Proactive Routing Protocols

Reactive Routing Protocols

Definition

Reactive routing protocols establish routes:

- Only when needed.

These are also called:

Working

1. Route discovery starts when data needs transmission.
2. Route maintained until communication ends.

Examples of Reactive Protocols

Protocol	Full Form
AODV	Ad Hoc On-Demand Distance Vector
DSR	Dynamic Source Routing
TORA	Temporally Ordered Routing Algorithm

Diagram of Reactive Routing

```
Need Route
      ↓
Route Discovery
      ↓
Data Transmission
```

Advantages of Reactive Protocols

1. Lower Routing Overhead

Routes are created only when needed.

Benefit

Less control traffic.

2. Better Bandwidth Utilization

No periodic routing updates required.

Benefit

Efficient use of wireless bandwidth.

---

3. Suitable for Highly Dynamic Networks

Routes adapt dynamically according to mobility.

---

4. Energy Efficient

Fewer routing messages reduce battery consumption.

Disadvantages of Reactive Protocols

---

1. Route Discovery Delay

Communication cannot start immediately because route discovery takes time.

Effect

Higher latency.

---

2. Flooding Overhead

Broadcasting route requests may increase congestion.

---

3. Poor Performance in Heavy Traffic

Frequent route discoveries reduce efficiency.

---

4. Route Maintenance Complexity

Broken routes require rediscovery.

Proactive Routing Protocols

Definition

Proactive routing protocols continuously maintain routing information for all nodes.

These are also called:

```
Table-Driven Routing Protocols
```

Working

1. Nodes periodically exchange routing updates.
2. Routing tables remain updated continuously.

Examples of Proactive Protocols

Protocol	Full Form
DSDV	Destination Sequenced Distance Vector
OLSR	Optimized Link State Routing
WRP	Wireless Routing Protocol

Diagram of Proactive Routing

```
Periodic Route Updates
          ↓
Routing Tables Maintained
          ↓
Immediate Data Transmission
```

Advantages of Proactive Protocols

---

1. Low Communication Delay

Routes are already available.

Benefit

Immediate packet transmission.

---

2. Better for Stable Networks

Works efficiently when topology changes slowly.

---

3. Easy Route Availability

No need for route discovery before transmission.

---

4. Consistent Network Information

Nodes maintain updated routing tables.

Disadvantages of Proactive Protocols

---

1. High Routing Overhead

Periodic updates consume:

- Bandwidth
- Battery power

---

2. Poor Scalability

Large networks create huge routing tables.

---

3. Wastage of Resources

Routes maintained even if never used.

---

4. High Energy Consumption

Frequent updates drain battery power.

Comparison Between Reactive and Proactive Protocols

Feature	Reactive Protocols	Proactive Protocols
Route Creation	On-demand	Continuous
Routing Overhead	Lower	Higher
Delay	Higher	Lower
Bandwidth Usage	Efficient	More consumption
Scalability	Better	Limited

Feature	Reactive Protocols	Proactive Protocols
Energy Consumption	Lower	Higher
Route Availability	Delayed	Immediate

Real-Life Example

Reactive Protocol Example

Military communication where:

- Nodes move rapidly.
- Routes change frequently.

Reactive protocols are preferred.

Proactive Protocol Example

Office wireless networks where:

- Topology remains relatively stable.

Proactive protocols work efficiently.

Applications of Routing Protocols

Application	Suitable Protocol
Disaster Recovery	Reactive
Battlefield Communication	Reactive
Campus Networks	Proactive
Sensor Networks	Proactive

Conclusion

Message routing in ad hoc wireless networks is challenging because of dynamic topology, limited bandwidth, mobility, energy constraints, and security threats. Routing protocols are broadly classified into reactive and proactive protocols. Reactive protocols establish routes only when needed, reducing routing overhead but increasing delay. Proactive protocols maintain continuous routing information, enabling fast communication but increasing overhead and resource consumption. Both approaches have advantages and disadvantages, and the choice depends on network size, mobility, and application requirements.

Q5c) Explain about global state routing protocol with suitable used cases.

Ans 5c) Global State Routing (GSR) Protocol

Introduction

In Mobile Ad Hoc Networks (MANETs), routing protocols are used to discover and maintain communication paths between mobile nodes. Because MANETs have:

- Dynamic topology
- Frequent node movement
- No fixed infrastructure

efficient routing becomes very important.

One important proactive routing protocol used in MANETs is:

```
Global State Routing (GSR)
```

GSR is designed to reduce routing overhead while maintaining updated routing information in highly dynamic wireless networks.

Definition of Global State Routing (GSR)

Global State Routing (GSR) is a proactive link-state routing protocol for mobile ad hoc networks in which each node maintains routing information about the entire network topology and updates routing tables periodically.

It is an improved version of:

```
Link State Routing
```

used in traditional networks.

Characteristics of GSR Protocol

1. Proactive routing protocol.
2. Table-driven routing approach.
3. Maintains topology information.
4. Supports dynamic mobile networks.
5. Reduces routing overhead compared to traditional link-state routing.

Need for GSR Protocol

Traditional routing protocols generate:

- Excessive flooding
- High control overhead

in MANET environments.

GSR was developed to:

1. Reduce unnecessary flooding.
2. Improve routing efficiency.
3. Support node mobility.
4. Maintain updated network topology.

Working Principle of GSR Protocol

In GSR:

- Each node maintains information about the complete network topology.
- Nodes exchange link-state information periodically with neighboring nodes.
- Routing tables are updated continuously.

Unlike traditional link-state protocols:

- GSR reduces flooding overhead.

Routing Information Maintained in GSR

Each node maintains several tables:

1. Neighbor List
2. Topology Table
3. Next Hop Table
4. Distance Table

---

1. Neighbor List

Purpose

Stores information about directly connected neighboring nodes.

Example

```
Node A → Node B, Node C
```

---

2. Topology Table

Purpose

Stores complete network topology information.

Functions

1. Maintains link-state updates.
2. Tracks network structure.

---

3. Next Hop Table

Purpose

Stores the next node used to reach destination.

Example

```
Destination D → Next Hop B
```

---

4. Distance Table

Purpose

Stores shortest path distance to all nodes.

Working of GSR Protocol

Step 1: Neighbor Discovery

Each node identifies neighboring nodes using periodic messages.

Step 2: Link State Exchange

Nodes exchange topology information with neighbors.

Step 3: Topology Update

Each node updates:

- Topology table
- Distance table
- Routing information

Step 4: Route Calculation

Shortest paths are calculated using routing algorithms.

Step 5: Data Transmission

Packets are forwarded using updated routing tables.

Diagram of GSR Working

```
Node Discovery
        ↓
Link-State Exchange
        ↓
Topology Update
        ↓
Shortest Path Calculation
        ↓
Data Routing
```

Example of GSR Operation

Suppose a MANET contains:

- Node A
- Node B
- Node C
- Node D

If Node A wants to send data to Node D:

1. Node A checks routing table.
2. Finds shortest path through intermediate nodes.
3. Data forwarded accordingly.

Example route:

```
A → B → D
```

Features of GSR Protocol

1. Maintains global topology information.
2. Uses shortest path routing.
3. Reduces flooding overhead.
4. Supports dynamic networks.

5. Provides loop-free routing.

Advantages of GSR Protocol

---

1. Reduced Routing Overhead

Unlike traditional link-state routing:

- GSR limits unnecessary flooding.

Benefit

Efficient bandwidth usage.

---

2. Faster Route Availability

Routes are already maintained proactively.

Benefit

Reduced communication delay.

---

3. Loop-Free Routing

Maintains accurate topology information.

Benefit

Reliable packet delivery.

---

4. Better Topology Awareness

Each node knows complete network structure.

Benefit

Efficient route calculation.

---

5. Supports Mobility

Handles dynamic topology changes effectively.

Disadvantages of GSR Protocol

---

1. High Storage Requirement

Each node stores:

- Global network topology information.

Problem

Increased memory usage.

---

2. Periodic Updates Required

Frequent updates consume:

- Bandwidth
- Battery power

---

3. Scalability Issues

Large networks increase:

- Routing table size
- Update complexity

---

4. Processing Overhead

Continuous topology calculations increase CPU usage.

Comparison with Traditional Link-State Routing

Feature	Traditional Link-State	GSR
Flooding	Extensive flooding	Reduced flooding
Routing Overhead	High	Lower
Mobility Support	Limited	Better
Scalability	Moderate	Better

Applications / Use Cases of GSR Protocol

---

1. Military Communication Networks

Usage

Used in battlefield communication where:

- Nodes move dynamically.
- Reliable routing is required.

Benefit

Provides updated routing information quickly.

---

2. Disaster Recovery Operations

Usage

Temporary communication networks during:

- Earthquakes
- Floods
- Emergency rescue operations

Benefit

Infrastructure-less communication support.

---

3. Vehicular Ad Hoc Networks (VANETs)

Usage

Communication between moving vehicles.

Benefit

Supports dynamic route management.

---

4. Wireless Sensor Networks

Usage

Sensor nodes communicate dynamically.

Benefit

Efficient topology management.

---

5. Campus or Conference Networks

Usage

Temporary mobile wireless networks in educational or conference environments.

Benefit

Fast routing setup without fixed infrastructure.

Comparison Between Proactive and Reactive Routing

Feature	GSR (Proactive)	Reactive Protocols
Route Availability	Immediate	On-demand
Delay	Low	Higher
Routing Overhead	Moderate	Lower
Scalability	Moderate	Better in large dynamic networks

Why GSR is Important in MANET

GSR helps:

1. Maintain updated routes.
2. Reduce routing delay.
3. Improve packet delivery.
4. Support mobility efficiently.

Real-Life Example

Suppose rescue teams communicate using MANET after a natural disaster:

- Team members move frequently.
- Network topology changes continuously.

GSR maintains updated routing information so communication continues smoothly.

Challenges in GSR Protocol

1. Large routing tables.
2. Frequent topology updates.
3. Battery consumption.
4. Bandwidth usage.

Summary of GSR Protocol

Aspect	Description
Type	Proactive Routing Protocol
Routing Method	Link-state based
Main Goal	Reduce flooding overhead
Best Suitable For	Dynamic MANET environments

Conclusion

Global State Routing (GSR) is a proactive routing protocol designed for Mobile Ad Hoc Networks. It maintains complete topology information at each node and continuously updates routing tables through periodic link-state exchanges. GSR reduces routing overhead compared to traditional link-state protocols while providing fast route availability and efficient packet delivery. Although it requires more memory and periodic updates, GSR is highly useful in dynamic wireless environments such as military communication, disaster recovery, and vehicular networks.